



AWS Academy Cloud Foundations Labs 1 to 3

By Albert Zhang

Background:

Lab 1:

In the first lab the user learns about **IAM (identity and access management)**:

An IAM Define fine-grained access rights:

- Who can access the resource
- Which resources can be accessed and what can the user do to the resource
- How resources can be accessed

(IAM is a no-cost AWS account feature)

IAM Authorization:

Authorization is the process of determining what permission a user, service or application should be granted. After user has been authenticated, they must be authorized to access AWS services

By default, IAM users do not have permission to access any resource or data in an AWS account. Instead, explicit permission needs to be granted to a user, group or role by creating a policy, which is a document in JSON format. A policy list permission that allows or denies access to resources in the AWS account.

This lab demonstrates how to access pre-created IAM Users and groups by inspecting IAM policies as applied to the pre-created groups, this lab also follows a real-world scenario, teaching the

Lab 2:

Amazon Virtual Private Cloud (Amazon VPC):

The Amazon VPC lets someone launch Amazon Web Services resources in a virtual network that is customizable. The virtual network is very similar to traditional network, with the difference being that Amazon VPC can be scalable using the infrastructure of AWS. VPCs can be created to cover multiple Availability Zones

The second lab shows the user how to use Amazon Virtual Private Cloud (VPC) to create their own VPC and add more components to produce a customized network. This lab also demonstrates how to create a security group and customize an EC2 instance which can run a web server, with the EC2 instance running a subnet in the VPC.

After someone completes this lab, they should be able to create a VPC, create subnets, configure a security group, and finally be able to launch an EC2 instance into a VPC.

Lab 3:

Amazon Elastic Computer Cloud (EC2):

EC2 is a web service that provides resizable computing capacity in cloud. EC2 is designed to make web-scale cloud computing easier for developers.

Amazon EC2 has a simple web service interface that lets you obtain and configure capacity easily. It provides you with complete control of your computing resources and lets you run on Amazon's proven computing environment. Amazon EC2 reduces the time needed to get and boot new server instances to minutes, enabling you to quickly scale capacity either up or down, as your needs change.

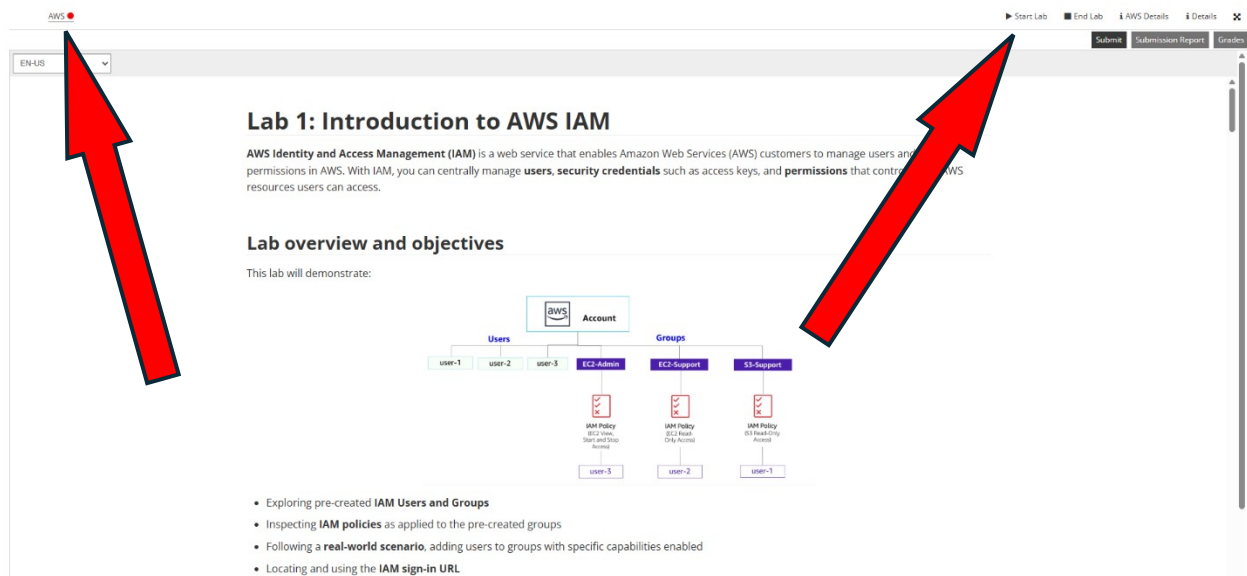
Amazon EC2 completely changes the economics of computing by allowing the user to pay only for capacity that one needs. Amazon EC2 gives developers the tools to build failure resilient applications and separate themselves from common failure scenarios

This lab provides the user with a basic overview of launching, resizing, managing and monitoring an Amazon EC2 instance. After completing the lab, one should be able to launch a web server with termination protection enabled, monitor your EC2 instance, modify security group that one's web server is using, resize Amazon EC2 instance to scale, enabling stop protection, and finally stopping the EC2 instance.

Configurations:

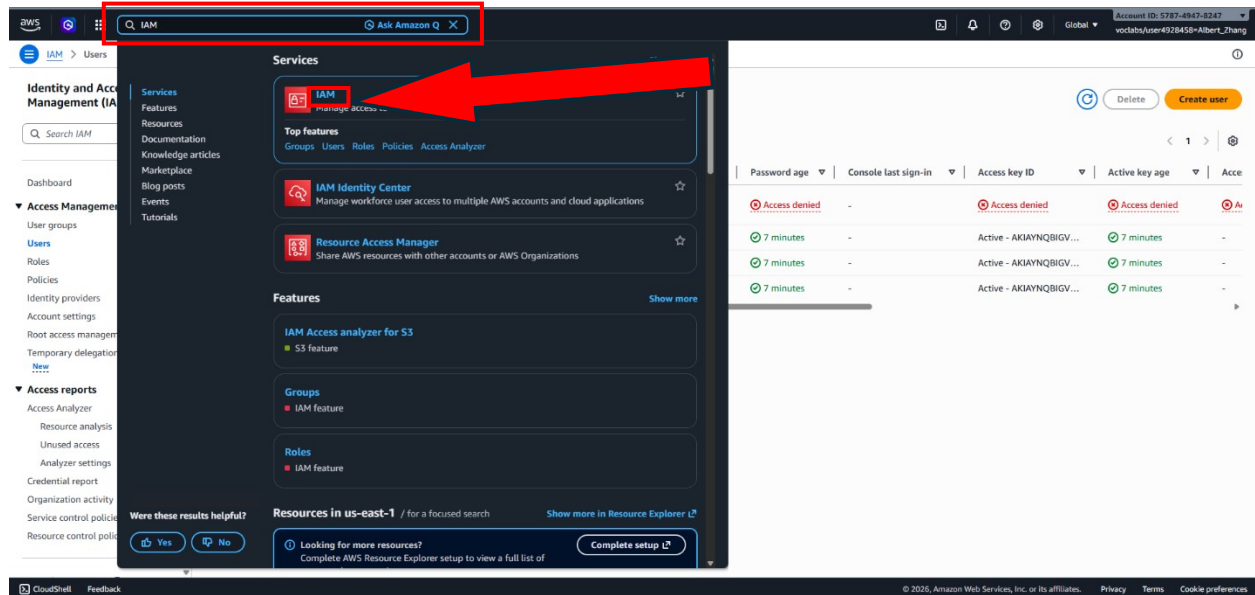
Lab 1:

Star with clicking "Start Lab" and then the AWS link

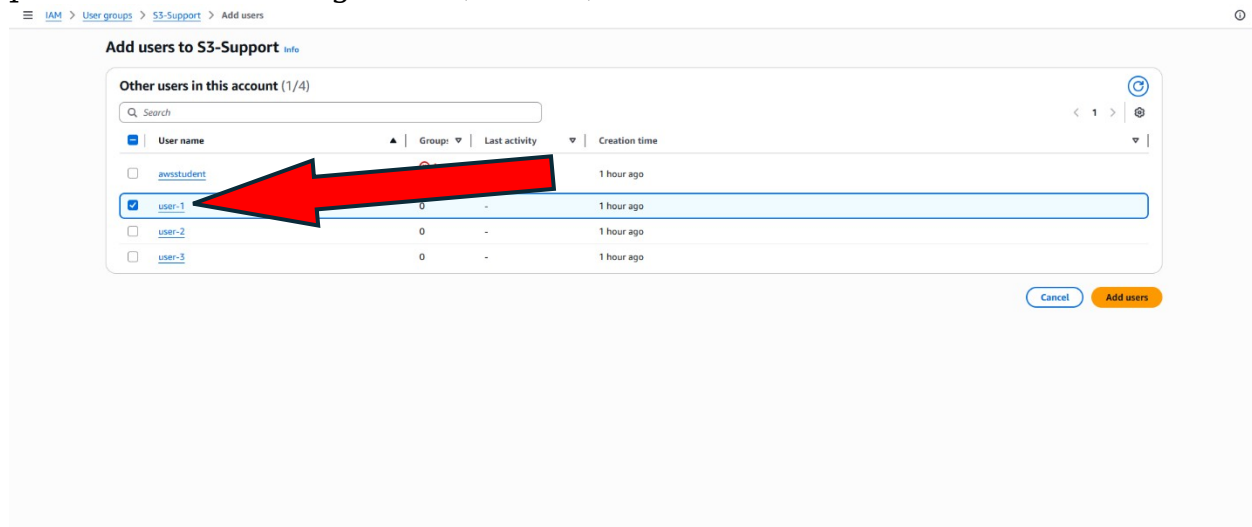


The screenshot shows the AWS IAM Lab 1 interface. At the top, there is a navigation bar with a dropdown menu set to 'EN-US'. On the right side of the navigation bar, there are buttons for 'Start Lab', 'End Lab', 'AWS Details', and 'Details'. Below the navigation bar, the main content area is titled 'Lab 1: Introduction to AWS IAM'. It includes a brief description of AWS Identity and Access Management (IAM) and a section titled 'Lab overview and objectives'. A diagram illustrates the IAM structure, showing an 'AWS Account' at the top, branching into 'Users' (user-1, user-2, user-3) and 'Groups' (EC2-Admin, EC2-Support, S3-Support). Each group is associated with an 'IAM Policy' (EC2-Admin, EC2-Support, S3-Support) and a 'user-1' instance. Below the diagram, there are four bullet points: 'Exploring pre-created IAM Users and Groups', 'Inspecting IAM policies as applied to the pre-created groups', 'Following a real-world scenario, adding users to groups with specific capabilities enabled', and 'Locating and using the IAM sign-in URL'. Two large red arrows are overlaid on the image: one pointing to the 'Start Lab' button in the top right corner, and another pointing to the 'AWS' link in the top left corner of the main content area.

Now you will be brought to the AWS page, go to the right of Services and search IAM to open the IAM console



Go to the navigate pane on the left and choose “Users”, you will see the 3 premade users being user-1, user-2, and user-3. Choose the user-1 link



After going through the user-1 link click “Permissions”

EC2-Support Info Delete

Summary Edit

User group name: EC2-Support | Creation time: March 23, 2026, 20:45 (UTC-07:00) | ARN: arn:aws:iam::578749478247:group/spl66/EC2-Support

Users **Permissions** Access Advisor

Permissions policies (1) Info Simulate Remove Add permissions

can attach up to 10 managed policies.

Filter by Type: All types

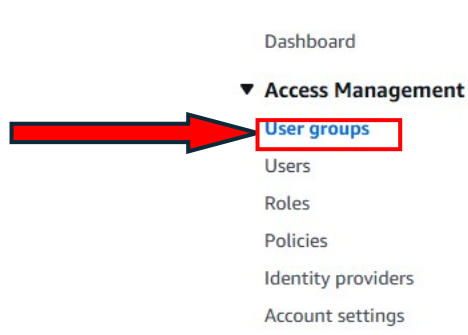
Policy name	Type	Attached entities
AmazonEC2ReadOnlyAccess	AWS managed	1

AmazonEC2ReadOnlyAccess Copy JSON

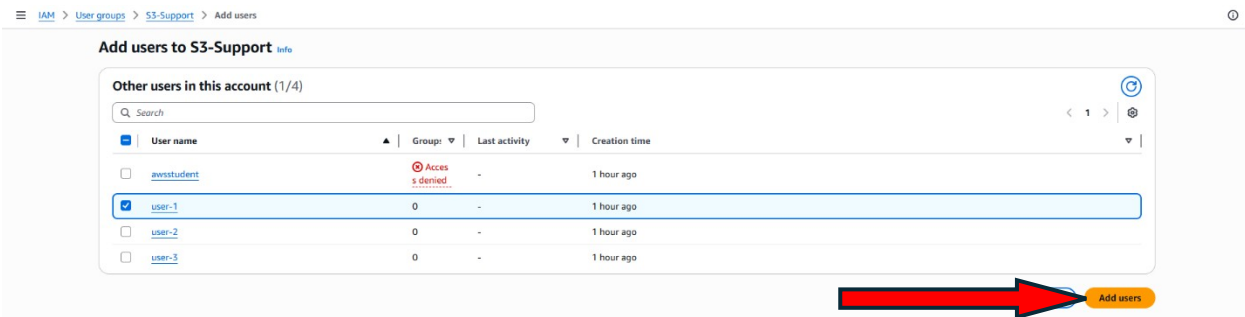
```
1- {
2-   "Version": "2012-10-17",
3-   "Statement": [
4-     {
5-       "Effect": "Allow",
6-       "Action": [
7-         "ec2:Describe*",
8-         "ec2:GetSecurityGroupsForVpc"
9-       ],
10-      "Resource": "*"
11-    },
12-    {
13-      "Effect": "Allow",
14-      "Action": "elasticloadbalancing:Describe"
```

You will see that this group is running something called “AmazonEC2ReadOnlyAccess”, click the plus(+) icon next to it to view the policy details. Repeat the same steps but for user-2 and 3.

Now you will add users to groups, go to the navigation pane again and choose “User groups this time.



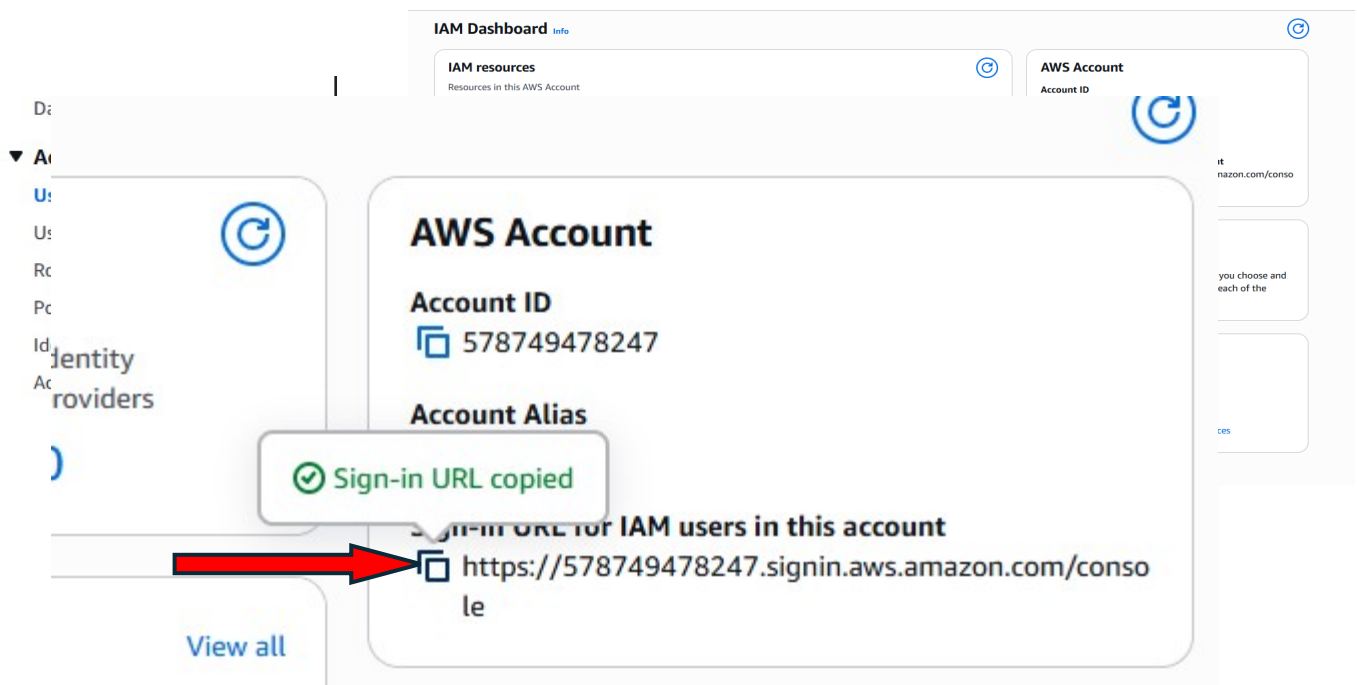
Choose the S3-Support group link and choose the Users tab, after that choose “Add users”, add user-1 to the group



Repeat the same steps for user-2 and user-3 but adding User-2 to the “EC2-Support Group” and user-3 to the “EC2-Admin Group”. When done successfully you will see this notification:



Next, you will sign-in and test users. On the navigation pane to the left, choose “Dashboard”, you will see a sign-in URL for IAM users in this account link is displayed on the right. The link can be used to sign-in to AWS account you are currently using.



Next, copy the sign-in URL for IAM users in this account

Open a new private window and paste the IAM user sign-in link you just copied. You will be brought to this page with your account ID or alias already entered in.

You will need to enter in the IAM username and password.

User-1:

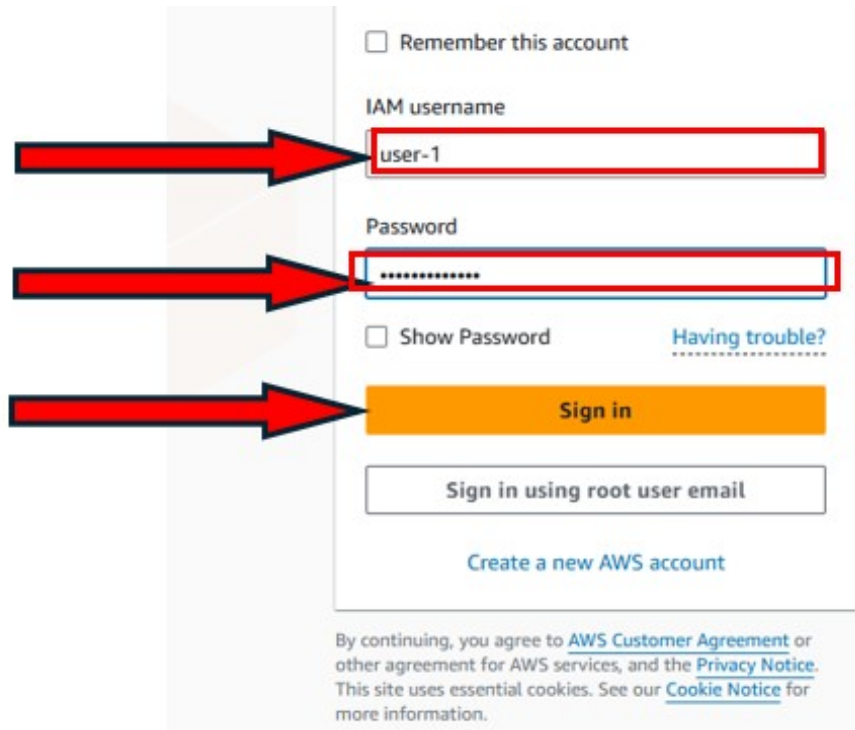
Username: user-1, password: Lab-Password1

User-2:

Username: user-2, password: Lab-Password2

User-3:

Username: user-3, password: Lab-Password3



The image shows the AWS IAM user sign-in page. It features a light gray background with a white sign-in box. At the top of the box is a checkbox labeled "Remember this account". Below it is the "IAM username" label followed by a text input field containing "user-1". A red arrow points from the left to this input field. Below the username field is the "Password" label followed by a password input field with masked characters. A second red arrow points from the left to this field. Below the password field is a checkbox labeled "Show Password" and a link labeled "Having trouble?". At the bottom of the sign-in box is an orange "Sign in" button, with a third red arrow pointing from the left to it. Below the sign-in box is a button labeled "Sign in using root user email" and a link labeled "Create a new AWS account". At the very bottom of the page, there is a small text block containing a disclaimer about the AWS Customer Agreement, Privacy Notice, and Cookie Notice.

☐ Remember this account

IAM username

user-1

Password

.....

☐ Show Password [Having trouble?](#)

Sign in

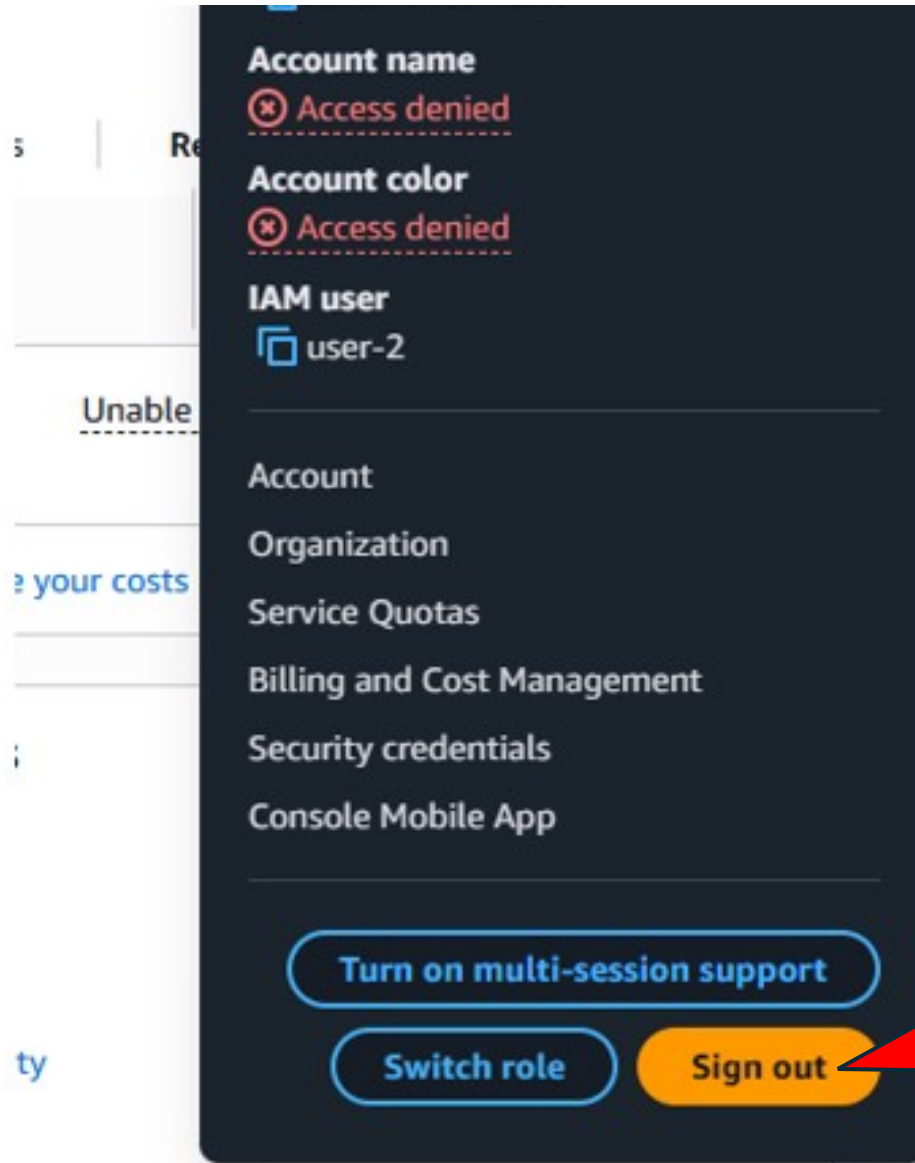
Sign in using root user email

[Create a new AWS account](#)

By continuing, you agree to [AWS Customer Agreement](#) or other agreement for AWS services, and the [Privacy Notice](#). This site uses essential cookies. See our [Cookie Notice](#) for more information.

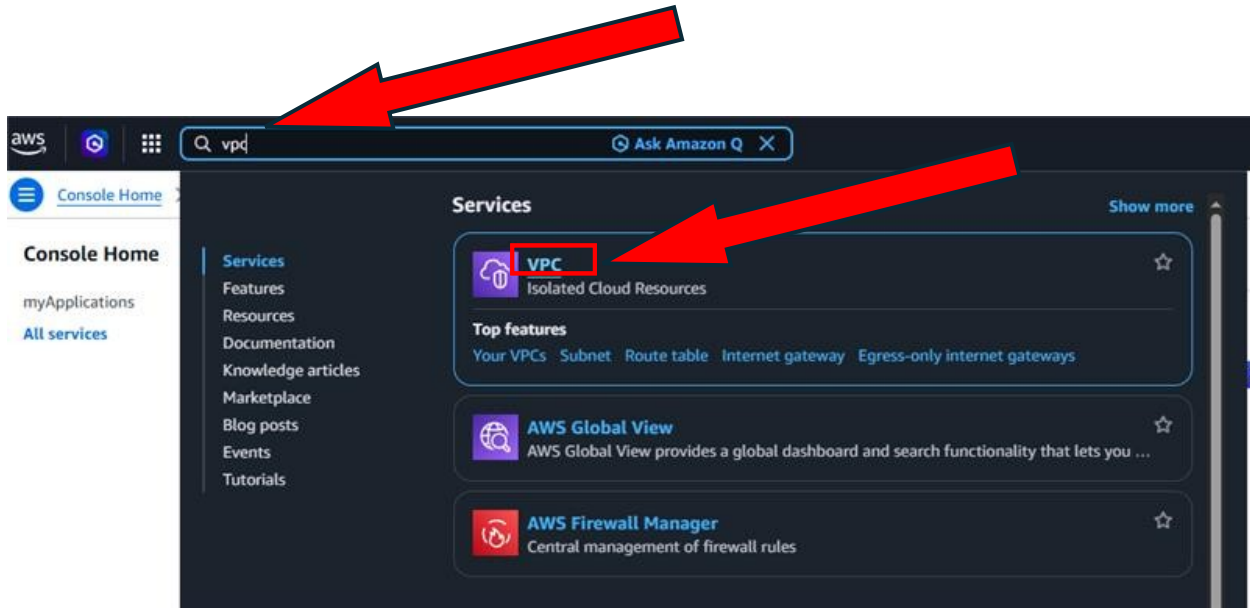
After signing in to the IAM user accounts, choose instances, and in the instance state menu select "Stop instance"

User-1: search for S3 in services, this account has permission to view a list of Amazon S3 buckets and the contents, after completing the steps go to the top of the screen, choose user-1, 2 or 3 and click "Sign out"

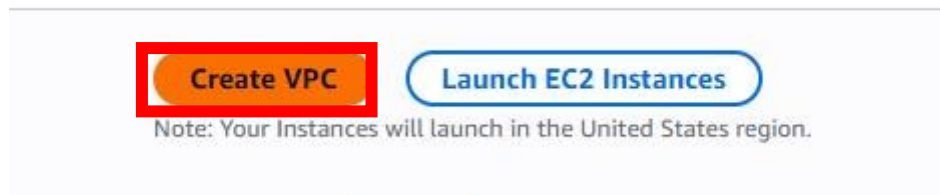


Lab 2:

In the search box to the right of Services search for “VPC” and open the VPC console



Next, begin creating a VPC. Select “Create VPC”



Choose VPC and more.

Under Name tag auto-generation, keep Auto-generate selected, however change the value from project to lab.

Keep the IPv4 CIDR block set to 10.0.0.0/16

For Number of Availability Zones, choose 1.

For Number of public subnets, keep the 1 setting.

For Number of private subnets, keep the 1 setting.

Expand the Customize subnets CIDR blocks section.

Change Public subnet CIDR block in us-east-1a to 10.0.0.0/24

Change Private subnet CIDR block in us-east-1a to 10.0.1.0/24

Set NAT gateways to

In 1 AZ.

Set VPC endpoints to None.

Keep both DNS hostnames and DNS resolution enabled.

Your configurations should look like:

VPC settings

Resources to create [Info](#)
Create only the VPC resource or the VPC and other networking resources.

☐ VPC only ☒ VPC and more

Name tag auto-generation [Info](#)
Enter a value for the Name tag. This value will be used to auto-generate Name tags for all resources in the VPC.

☒ Auto-generate
lab

IPv4 CIDR block [Info](#)
Determine the starting IP and the size of your VPC using CIDR notation.

10.0.0.0/16 65,536 IPs
CIDR block size must be between /16 and /28.

Pv6 CIDR block [Info](#)
☒ No IPv6 CIDR block
☐ Amazon-provided IPv6 CIDR block

Tenancy [Info](#)
Default

► **Encryption settings - optional**

Number of Availability Zones (AZs) [Info](#)
Choose the number of AZs in which to provision subnets. We recommend at least two AZs for high availability.

1 | 2 | 3

► **Customize AZs**

Number of public subnets [Info](#)
The number of public subnets to add to your VPC. Use public subnets for web applications that need to be publicly accessible over the internet.

0 | 1

Number of private subnets [Info](#)
The number of private subnets to add to your VPC. Use private subnets to secure backend resources that don't need public access.

0 | 1 | 2

▼ **Customize subnets CIDR blocks**

Public subnet CIDR block in us-east-1a
10.0.0.0/24 256 IPs

Private subnet CIDR block in us-east-1a
10.0.1.0/24 256 IPs

NAT gateways (\$) - updated [Info](#)

After that select "Create VPC" at the bottom of the screen:

NAT gateways (\$) - *updated* [Info](#)

NAT gateway allows private resources to access the internet from any availability zone within a VPC, providing a single managed internet exit point for the entire region. Additional charges apply.

None

 |

Regional - new

 |

Zonal

NAT gateways (\$) [Info](#)

Choose the number of Availability Zones (AZs) in which to create NAT gateways. Note that there is a charge for each NAT gateway

In 1 AZ

 |

1 per AZ

VPC endpoints [Info](#)

Endpoints can help reduce NAT gateway charges and improve security by accessing S3 directly from the VPC. By default, full access policy is used. You can customize this policy at any time.

None

 |

S3 Gateway

DNS options [Info](#)

☒ Enable DNS hostnames

☒ Enable DNS resolution

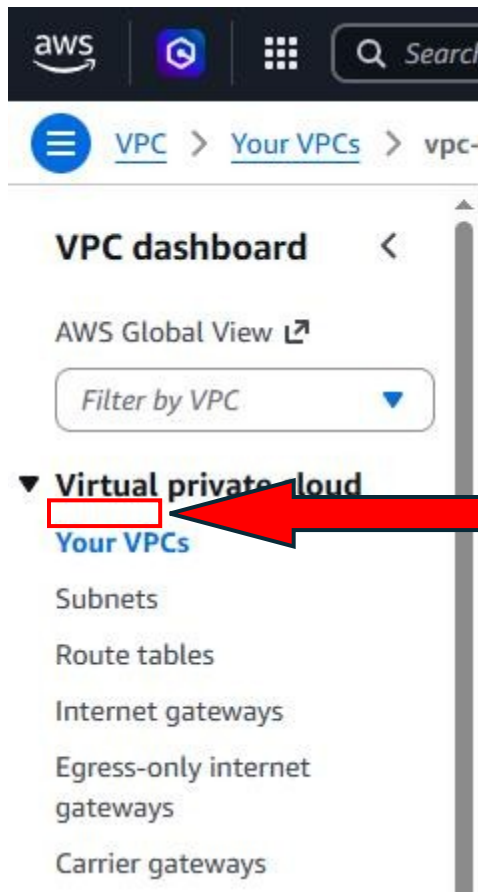
► **Additional tags**

Cancel

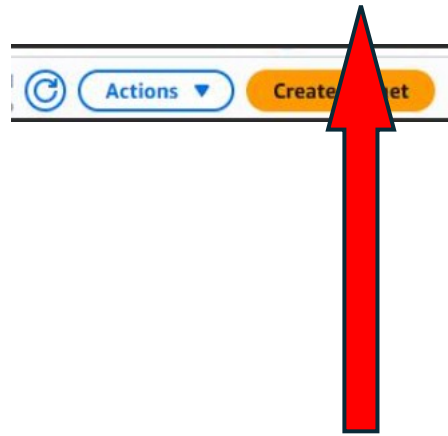
 **Preview code**

Create VPC

Next you will create a subnet, on the navigation pane, select “Subnets”



Select "Create Subnet", next to Actions



Now you will configure 2 new subnets with the configurations:

VPC ID: lab-vpc (select from the menu).

Subnet name: lab-subnet-public2

Availability Zone: Select the second Availability Zone (for example, us-east-1b)

IPv4 CIDR block: 10.0.2.0/24

The screenshot shows the 'Create subnet' page in the AWS Management Console. Red boxes and arrows highlight specific fields: a vertical arrow points to the 'VPC ID' field (containing 'vpc-034c0fa6968bba19e'), a horizontal arrow points to the 'Subnet name' field (containing 'lab-subnet-public2'), a horizontal arrow points to the 'Availability Zone' dropdown (set to 'us-east-1b'), and a horizontal arrow points to the 'IPv4 subnet CIDR block' field (containing '10.0.2.0/24'). The 'Create subnet' button at the bottom right is also highlighted with a red box.

Create subnet [Info](#)

VPC

VPC ID
vpc-034c0fa6968bba19e

Associated VPC CIDRs
IPv4 CIDRs
10.0.0.0/16

Subnet settings
Specify the CIDR blocks and the Availability Zone for the subnet.

Subnet 1 of 1

Subnet name
Create a tag with a key of `Name` and a value that you specify.
lab-subnet-public2
The name can be up to 255 characters long.

Availability Zone [Info](#)
Choose the region in which your subnet will reside, or let Amazon choose one for you.
United States (N. Virginia) / us-east-1b

IPv4 VPC CIDR block [Info](#)
Choose the VPC's IPv4 CIDR block for the subnet. The subnet's IPv4 CIDR must lie within this block.
10.0.0.0/16

IPv4 subnet CIDR block
10.0.2.0/24 256 IP

▼ Tags - optional

Key	Value - optional
Q Name	Q lab-subnet-public2

[Add new tag](#)
You can add 49 more tags.

[Remove](#)

[Add new subnet](#)

[Cancel](#) [Create subnet](#)

Followed by the next subnet:

VPC ID: lab-vpc

Subnet name: lab-subnet-private2

Availability Zone: Select the *second* Availability Zone (for example, us-east-1b)

IPv4 CIDR block: 10.0.3.0/24

Create subnet [Info](#)

VPC
VPC ID
 Create subnets in this VPC.
 vpc-054c0fa6968ba19e (lab-vpc) [▼](#)

Associated VPC CIDRs
 IPv4 CIDRs
 10.0.0.0/16

Subnet settings
 Specify the CIDR blocks and other settings for the subnet.

Subnet 1 of 1

Subnet name
 Create a tag with a key of "Name" and a value that you specify.
 lab-subnet-private2

Availability Zone [Info](#)
 Choose the zone in which your subnet will reside, or let Amazon choose one for you.
 United States (N. Virginia) / us-east-1a (us-east-1b)

IPv4 VPC CIDR block [Info](#)
 Choose the VPC's IPv4 CIDR block for the subnet. The subnet's IPv4 CIDR must be within this block.
 10.0.0.0/16

IPv4 subnet CIDR block
 10.0.3.0/24 (750 IPs)

Tags - optional
 Key: Name Value: lab-subnet-private2 [Remove](#)

[Add new tag](#) [Remove](#) [Add new subnet](#)

[Cancel](#) [Create subnet](#)

At the left navigation pane choose “Route Tables”, select lab-rtb-private1-us-east-1a

Route tables (1/6) [Info](#)

Name	Route table ID	Explicit subnet assoc...	Edge associations	Main	VPC	Owner ID
lab-rtb-private1-us-east-1a	rtb-0a34e2863ef201933	Subnet: subnet-0b80d9f1c0a8b9375	No	No	vpc-054c0fa6968ba19e (lab-vpc)	488514305041
lab-rtb-public	rtb-024888a0a758b3327	Subnet: subnet-073ab5c194701ee...	No	No	vpc-054c0fa6968ba19e (lab-vpc)	488514305041
Work Public Route Table	rtb-0a6f8c1a31a4fcaad	Subnet: subnet-07973c0a6a7d4e...	No	No	vpc-054c0fa6968ba19e (lab-vpc)	488514305041
	rtb-080bc142293c7356d		No	No	vpc-054c0fa6968ba19e (lab-vpc)	488514305041

lab-rtb-private1-us-east-1a [Details](#) [Routes](#) [Subnet associations](#) [Edge associations](#) [Route propagation](#) [Tags](#)

Explicit subnet associations (1) [Edit subnet associations](#)

Name	Subnet ID	IPv4 CIDR	IPv6 CIDR
lab-subnet-private1-us-east-1a	subnet-0b80d9f1c0a8b9375	10.0.3.0/24	--

Subnets without explicit associations (2) [Edit subnet associations](#)

The following subnets have not been explicitly associated with any route tables and are therefore associated with the main route table:

Name	Subnet ID	IPv4 CIDR	IPv6 CIDR
lab-subnet-public2	subnet-0e0f5ad67129133a6	10.0.2.0/24	--
lab-subnet-private2	subnet-0b33a3e3d3d6f022f	10.0.5.0/24	--

In explicit subnet associations panel, choose “Edit subnet associations”

Edit subnet associations

Change which subnets are associated with this route table.

Name	Subnet ID	IPv4 CIDR
lab-subnet-public2	subnet-0ec49fadc722933de	10.0.2.0/24
lab-subnet-public1-us-east-1a	subnet-07da60cf84780acc5	10.0.0.0/24
lab-subnet-private2	subnet-03c3c82fc696f052f	10.0.3.0/24
lab-subnet-private1-us-east-1a	subnet-0880491c09d85b7f5	10.0.1.0/24

Selected subnets

subnet-0880491c09d85b7f5 / lab-subnet-private1-us-east-1a X subnet-03c3c82fc696f052f / lab-subnet-private2 X

Cancel Save associations

Keep lab-subnet-private-us-east-1a selected and also select "lab-subnet-private2". Choose "Save associations"

Now you will create a VPC security group, acting like a firewall.

On the left navigation pane choose "Security groups", then choose "Create security groups" and then configure:

Security group name: web Security Group

Description: Enable HTTP access

VPC: choose the x to remove the currently selected VPC, then from the drop-down list choose "lab-vpc"

It should look something like:

Create security group [info](#)

A security group acts as a virtual firewall for your instance to control inbound and outbound traffic. To create a new security group, complete the fields below.

Basic details

Security group name [info](#)

Web Security Group
Name cannot be edited after creation.

Description [info](#)

Enable HTTP access

VPC [info](#)

vpc-034c0fa6968bba19e (lab-vpc)

Inbound rules [info](#)

Type [info](#)

HTTP

Protocol [info](#)

TCP

Port range [info](#)

80

Source [info](#)

Anywhere...

Description - optional [info](#)

Permit web requests

[Add rule](#)

Rules with source of 0.0.0.0/0 or ::/0 allow all IP addresses to access your instance. We recommend setting security group rules to allow access from known IP addresses only.

Outbound rules [info](#)

Type [info](#)

All traffic

Protocol [info](#)

All

Port range [info](#)

All

Destination [info](#)

Custom

Description - optional [info](#)

[Add rule](#)

Rules with destination of 0.0.0.0/0 or ::/0 allow your instances to send traffic to any IPv4 or IPv6 address. We recommend setting security group rules to be more restrictive and to only allow traffic to specific known IP addresses.

Tags - optional

A tag is a label that you assign to an AWS resource. Each tag consists of a key and an optional value. You can use tags to search and filter your resources or track your AWS costs.

No tags associated with the resource.

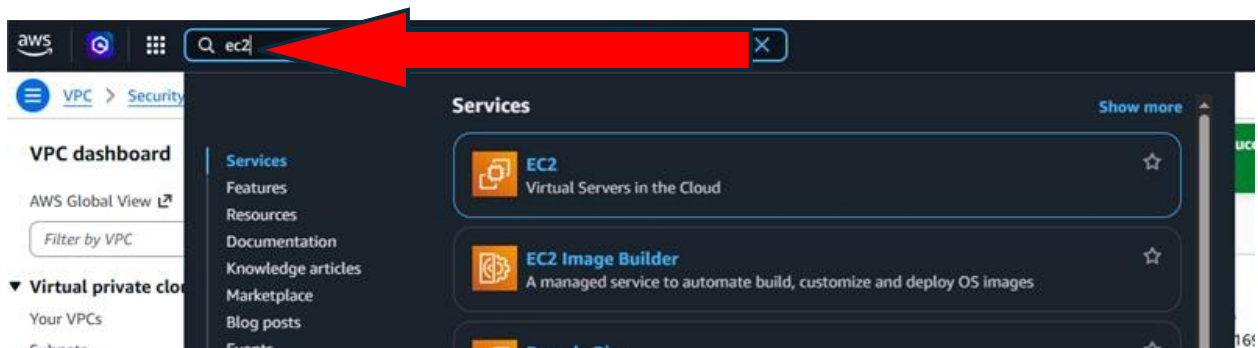
[Add new tag](#)

You can add up to 50 more tags.

[Cancel](#)

[Create security group](#)

Now you will launch an Amazon EC2 instance into the new VPC. In Services search for and choose EC2 and open the EC2 console



From the launch instance menu choose "Launch instance", give the instance the name Web Server 1.

Select the interface type t2.micro.

Launch an instance [Info](#)

Amazon EC2 allows you to create virtual machines, or instances, that run on the AWS Cloud. Quickly get started by following the simple steps below.

Name and tags [Info](#)

Name

Web Server 1

[Add additional tags](#)

▼ Application and OS Images (Amazon Machine Image) [Info](#)

An AMI contains the operating system, application server, and applications for your instance. If you don't see a suitable AMI below, use the search field or choose [Browse more AMIs](#).

Recents

Quick Start



[Browse more AMIs](#)
Including AMIs from
AWS, Marketplace and
the Community

Amazon Machine Image (AMI)

Amazon Linux 2023 kernel-6.1 AMI
ami-02dfbd4ff395f2a1b (64-bit (x86), uefi-preferred) / ami-0b11e0ed3f8697f97 (64-bit (Arm), uefi)
Virtualization: hvm ENA enabled: true Root device type: ebs

Description

Amazon Linux 2023 (kernel-6.1) is a modern, general purpose Linux-based OS that comes with 5 years of long term support. It is optimized for AWS and designed to provide a secure, stable and high-performance execution environment to develop and run your cloud applications.

Amazon Linux 2023 AMI 2023.10.20260302.1 x86_64 HVM kernel-6.1

Architecture

64-bit (x86)

Boot mode

uefi-preferred

AMI ID

ami-02dfbd4ff395f2a1b

Publish Date

2026-03-03

Username

ec2-user



Verified provider

▼ Instance type [Info](#) | [Get advice](#)

Instance type

t2.micro
Family: t2 1 vCPU 1 GiB Memory Current generation: true On-Demand Linux base pricing: 0.0116 USD per Hour
On-Demand Windows base pricing: 0.0162 USD per Hour On-Demand Ubuntu Pro base pricing: 0.0134 USD per Hour
On-Demand SUSE base pricing: 0.0116 USD per Hour On-Demand RHEL base pricing: 0.026 USD per Hour

☐ All generations

[Compare instance types](#)

Additional costs apply for AMIs with pre-installed software

From the key pair name menu select “vockey” and choose “Edit” then configure”:

Network: lab-vpc

Subnet: lab-subnet-public2

Auto-assign public IP: Enable

Under firewall, choose “Select existing security group”

Under common security groups, choose “Web Security Group”

It should look like:

The screenshot shows the AWS Management Console configuration page for an EC2 instance. Red arrows highlight the following settings:

- Key pair (login):** The 'Key pair name' dropdown is set to 'vockey'.
- Network settings:**
 - The 'VPC' dropdown is set to 'vpc-034c0fa6968bba19e (lab-vpc)'.
 - The 'Subnet' dropdown is set to 'subnet-0ec495adc722933de lab-subnet-public2'.
 - The 'Auto-assign public IP' dropdown is set to 'Enable'.
 - Under 'Firewall (security groups)', the 'Select existing security group' radio button is selected.
 - In the 'Common security groups' list, 'Web Security Group' is selected.

At the bottom, the 'Configure storage' section is visible, showing '1x 8 GiB gp3' storage configuration.

Expand the “Advanced details” panel
And paste this code into the user data box:

```
#!/bin/bash
# Install Apache Web Server and PHP
dnf install -y httpd wget php mariadb105-server
# Download Lab files
wget https://aws-tc-largeobjects.s3.us-west-2.amazonaws.com/CUR-TF-100-ACCLFO-2/2-lab2-vpc/s3/lab-app.zip
unzip lab-app.zip -d /var/www/html/
# Turn on web server
chkconfig httpd on
service httpd start
```

V2 only (token required)

For V2 requests, you must include a session token in all instance metadata requests. Applications or agents that use V1 for instance metadata access will break.

Metadata response hop limit [Info](#)
2

Allow tags in metadata [Info](#)
Select

User data - optional [Info](#)
Upload a file with your user data or enter it in the field.
[Choose file](#)

```
#!/bin/bash
# Install Apache Web Server and PHP
sudo yum install -y httpd wget php mariadb105-server
# Download Lab files
wget https://aws-tc-largeobjects.s3-us-west-2.amazonaws.com/CUR-TF-100-ACCLFO-2/2-lab2-vpc/S3/lab-app.zip
unzip lab-app.zip -d /var/www/html/
# Turn on web server
chkconfig httpd on
service httpd start
```


☐ User data has already been base64 encoded

Free tier: In your first year of opening an AWS account, you get 750 hours per month of t2.micro instance usage (or t3.micro where t2.micro isn't available) when used with free tier AMIs, 750 hours per month of public IPv4 address usage, 30 GiB of EBS storage, 2 million I/Os, 1 GiB of snapshots, and 100 GiB of bandwidth to the internet. Data transfer charges are not included as part of the free tier allowance. Charges may apply depending on your account's free tier status.

Cancel

Launch Instance

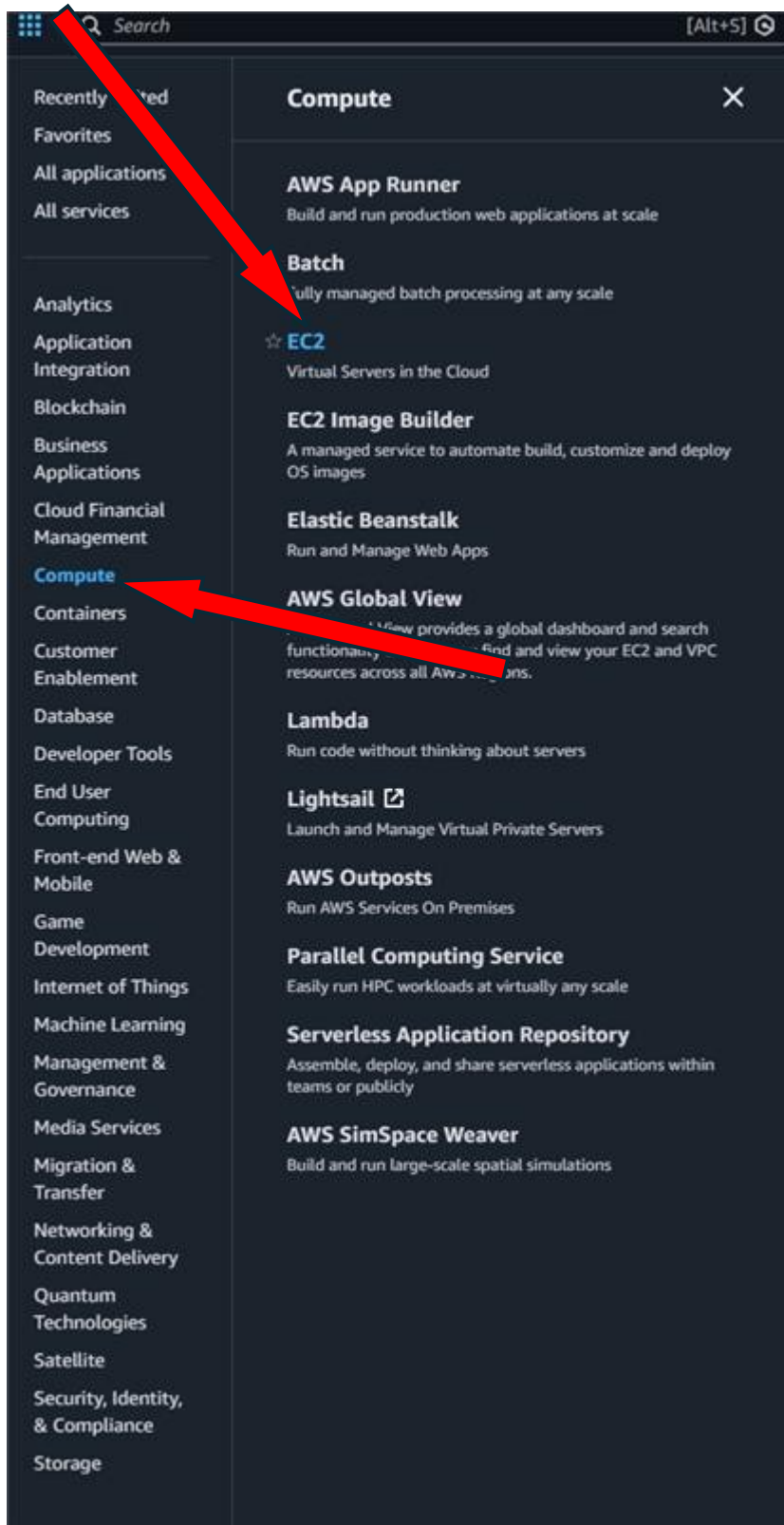
 [Preview code](#)



Finally, click “Lauch Instance”

Lab 3:

Go to services and choose “Compute” and choose “EC2”



After that, choose “Launch instance”

Launch instance

To get started, launch an Amazon EC2 instance, which is a virtual server in the cloud.

Launch instance

Create a server

Note: Your instances will launch in the United States (N. Virginia) Region

Set the instance name as "Web Server"

Launch an instance

Amazon EC2 allows you to create virtual machines, or instances, that run on the AWS Cloud. Quickly get started by following the simple steps below.

Name and tags

Name

Web Server

Add additional tags

Application and OS Images (Amazon Machine Image)

An AMI contains the operating system, application server, and applications for your instance. If you don't see a suitable AMI below, use the search field or choose [Browse more AMIs](#).

Search our full catalog including 1000s of application and OS images

Recents

Quick Start



[Browse more AMIs](#)
Including AMIs from
AWS, Marketplace and
the Community

Amazon Machine Image (AMI)

Amazon Linux 2023 kernel-6.1 AMI
ami-02dfbd4ff395f2a1b (64-bit (x86), uefi-preferred) / ami-02dfbd4ff395f2a1b (64-bit (Arm), uefi)
Virtualization: hvm ENA enabled: true Root device type: ebs

Description

Amazon Linux 2023 (kernel-6.1) is a modern, general purpose Linux-based OS that comes with 5 years of long term support. It is optimized for AWS and designed to provide a secure, stable and high-performance execution environment to develop and run your cloud applications.

Amazon Linux 2023 AMI 2023.10.20260302.1 x86_64 HVM kernel-6.1

Architecture	Boot mode	AMI ID	Publish Date	Username
64-bit (x86)	uefi-preferred	ami-02dfbd4ff395f2a1b	2026-03-03	ec2-user

Verified provider

Instance type

Instance type

t2.micro
Family: t2 1 vCPU 1 GiB Memory Current generation: true On-Demand Linux base pricing: 0.0116 USD per Hour
On-Demand Windows base pricing: 0.0162 USD per Hour On-Demand Ubuntu Pro base pricing: 0.0134 USD per Hour
On-Demand SUSE base pricing: 0.0116 USD per Hour On-Demand RHEL base pricing: 0.026 USD per Hour

All generations

[Compare instance types](#)

Additional costs apply for AMIs with pre-installed software

In the Quick Start AMIs, keep default Amazon Linux AMI selected.

In the Key pair login section, choose “vockey” for the Key pair name

The screenshot shows the AWS Management Console configuration page for an EC2 instance. Red arrows highlight specific configuration steps:

- Key pair (login)**: The "Key pair name - required" dropdown is set to "vockey".
- Network settings**: The "Auto-assign public IP" option is set to "Enable".
- Firewall (security groups)**: The "Create security group" button is selected, and the "Security group name - required" field is set to "Web Server security group".

Other visible configuration details include:

- VPC**: vpc-0aac8bd58a68c5791 (Lab VPC)
- Subnet**: subnet-00881d5ec9e5276f7 (PublicSubnet1)
- Description**: Security group for new web server
- Configure storage**: 1x 8 GiB gp3 Root volume, 3000 IOPS, Not encrypted

Expand and then go to “Advanced Details”. Select enable for Termination protection

▼ Advanced details [Info](#)

Domain join directory [Info](#)

Select [Create new directory](#)

IAM instance profile [Info](#)

Select [Create new IAM profile](#)

Hostname type [Info](#)

IP name

DNS Hostname [Info](#)

☒ Enable IP name IPv4 (A record) DNS requests

☐ Enable resource-based IPv4 (A record) DNS requests

☐ Enable resource-based IPv6 (AAAA record) DNS requests

Instance auto-recovery [Info](#)

Select

Shutdown behavior [Info](#)

Stop

Stop - Hibernate behavior [Info](#)

Select

Termination protection [Info](#)

Select

Select ✓

Enable

Disable

Detailed CloudWatch monitoring [Info](#)

Select

Credit specification [Info](#)

Standard

Placement group [Info](#)

Select [Create new placement group](#)

Go all the way to the bottom of the page and copy this code into the “User data” box:

```
#!/bin/bash
dnf install -y httpd
systemctl enable httpd
systemctl start httpd
echo '<html><h1>Hello From Your Web Server!</h1></html>' >
/var/www/html/index.html
```

It will look something like:

User data - optional [Info](#)

Upload a file with your user data or enter it in the field.

[↑ Choose file](#)

```
#!/bin/bash
dnf install -y httpd
systemctl enable httpd
systemctl start httpd
echo '<html><h1>Hello From Your Web Server</h1></html>' > /var/www/html/index.html
```

☐ User data has already been base64 encoded

After that, at the bottom of the “Summary” panel choose “Launch Instance”, if everything is done correctly, you will see a green success message at top of the page.

Success
Successfully initiated launch of instance i-056af37c1af071ee5d

Launch log

Next Steps
What would you like to do next with this instance, for example "create alarm" or "create backup" 1 2 3 4

Create billing and free tier usage alerts To manage costs and avoid surprise bills, set up email notifications for billing and free tier usage thresholds. Create billing alerts	Connect to your instance Once your instance is running, log into it from your local computer. Connect to instance Learn more	Connect an RDS database Configure the connection between an EC2 instance and a database to allow traffic flow between them. Connect an RDS database Create a new RDS database Learn more	Create EBS snapshot policy Create a policy that automates the creation, retention, and deletion of EBS snapshots. Create EBS snapshot policy	Manage detailed monitoring Enable or disable detailed monitoring for the instance. If you enable detailed monitoring, the Amazon EC2 console displays monitoring graphs with a 1-minute period. Manage detailed monitoring	Create Load Balancer Create a application, network gateway or classic Elastic Load Balancer. Create Load Balancer
Create AWS budget AWS Budgets allow you to create budgets, forecast spend, and take action on your costs and usage from a single location. Create AWS budget	Manage CloudWatch alarms Create or update Amazon CloudWatch alarms for the instance. Manage CloudWatch alarms	Disaster recovery for your instances Recover the instance you just launched into a different Availability Zone or a different Region using AWS Elastic Disaster Recovery (DRS). Disaster recovery for your instances	Monitor for suspicious runtime activities Amazon GuardDuty enables you to continuously monitor for malicious runtime activity and unauthorized behavior, with near real-time visibility into on-host activities occurring across your Amazon EC2 workloads. Monitor for suspicious runtime activities	Get instance screenshot Capture a screenshot from the instance and view it as an image. This is useful for troubleshooting on unreachable instances. Get instance screenshot	Get system log View the instance's system log to troubleshoot issues. Get system log

[View all instances](#)

Next, you will monitor your instance, go to the “Monitoring” tab, select enlarge. Then in the “Actions” menu on the top of the console, select “Monitor and troubleshoot” then go to “Get system log.”

The screenshot shows the AWS Management Console 'Instances' page. The table lists two instances: 'Web Server' (i-056a42e1af501ace2) and 'Backend Host' (i-09f57b04249054fe). The 'Web Server' instance is in a 'Running' state. A red arrow points to the 'Monitor and troubleshoot' link in the 'Actions' dropdown menu for the 'Web Server' instance. Another red arrow points to the 'Get system log' option in the same menu.

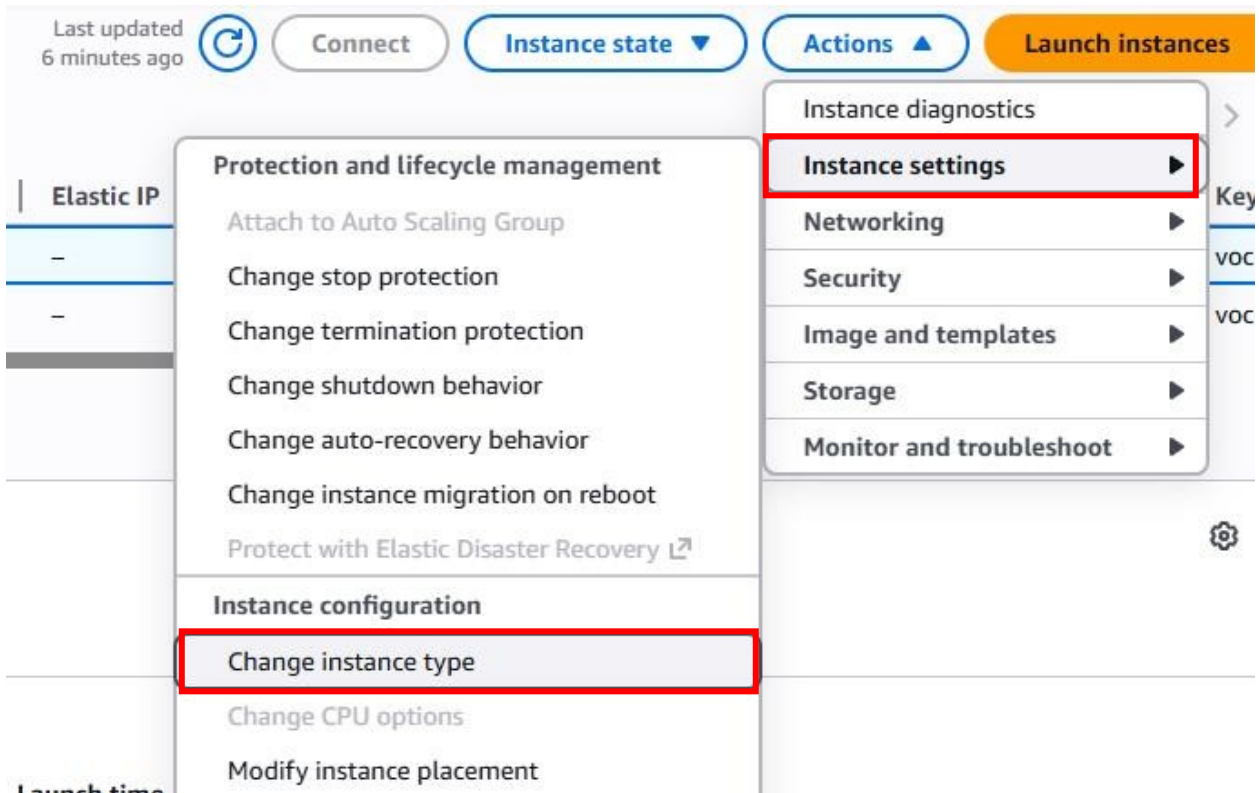
You will then see the user data that you added when creating the instance, through the system log, as shown below:

The screenshot shows the 'Instance diagnostics' page for the 'Web Server' instance. The 'System log' tab is selected, displaying the output of the 'cloud-init' command. The log shows the creation of a symlink and the generation of a random string. A red arrow points to the 'Connect' button at the bottom of the page.

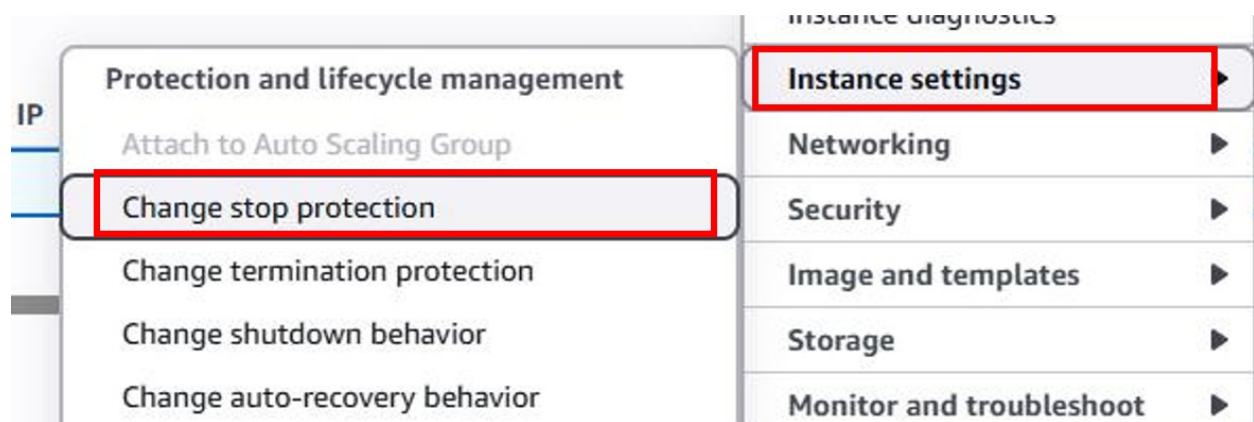
Now you will resize your instance, first stop your instance.

Now you will change the instance type and enable stop protection. Select the "Web Server" interface, and then in the "Actions" Menu, select "Instance setting", then "Change instance type", then configure the instance type to t2.small. Finally click "Apply".

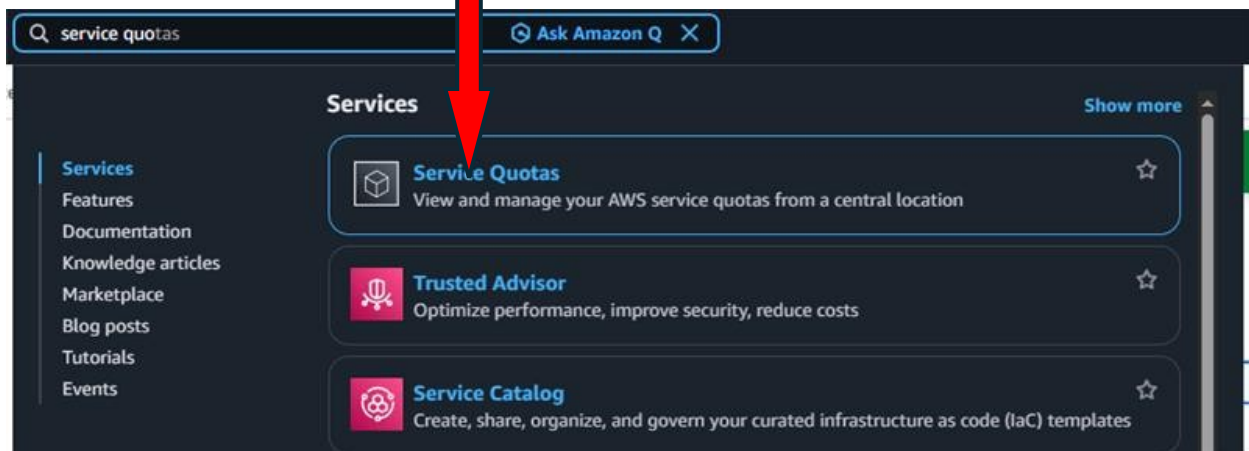
The screenshot shows the AWS Management Console 'Instances' page. The 'Web Server' instance is selected. A red arrow points to the 'Instance settings' link in the 'Actions' dropdown menu.



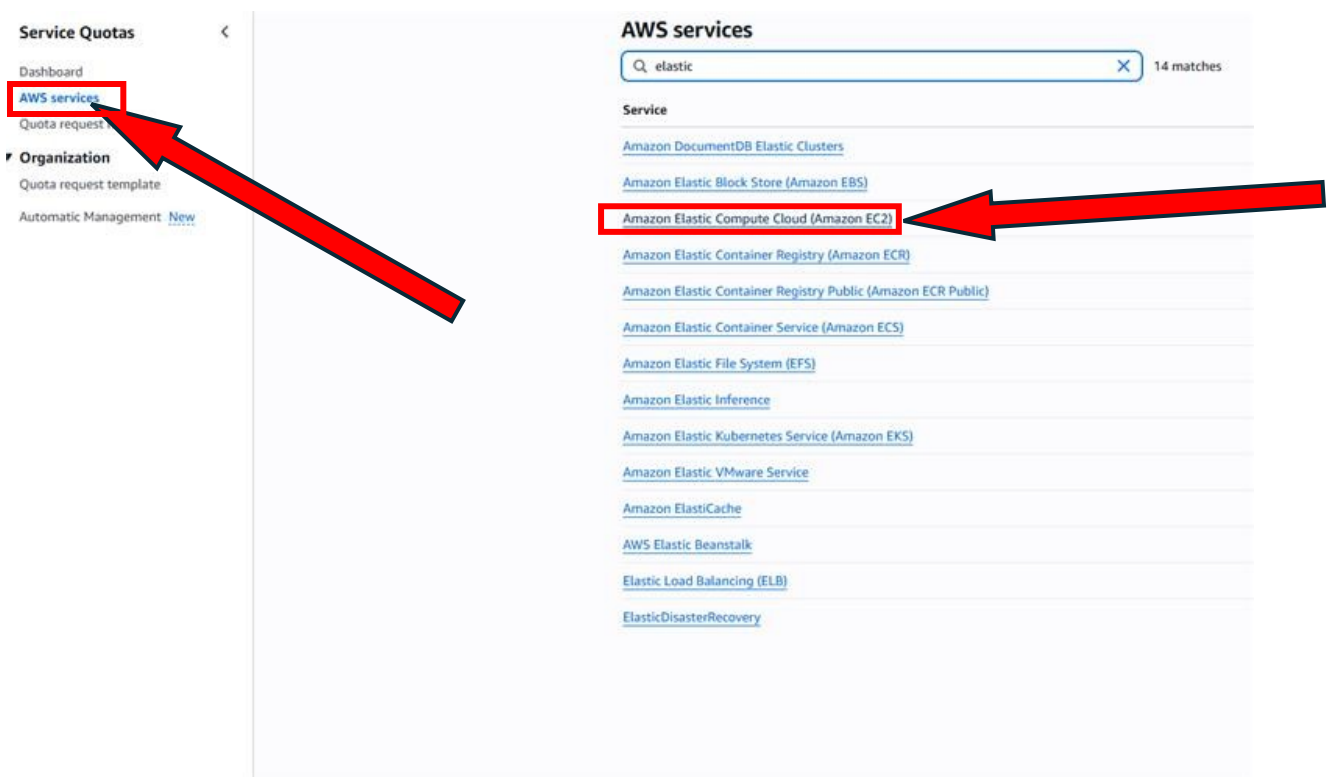
Select the “Web Service” interface again, and go to Actions, and select “Instances settings”, then “Change stop protection”



Next you will explore EC2 limits, in the box next to Services, search and select “Service Quotas”, choose “AWS services”



After Selecting Service Quotas, search for “Amazon Elastic Compute Cloud (Amazon EC2)” in AWS services

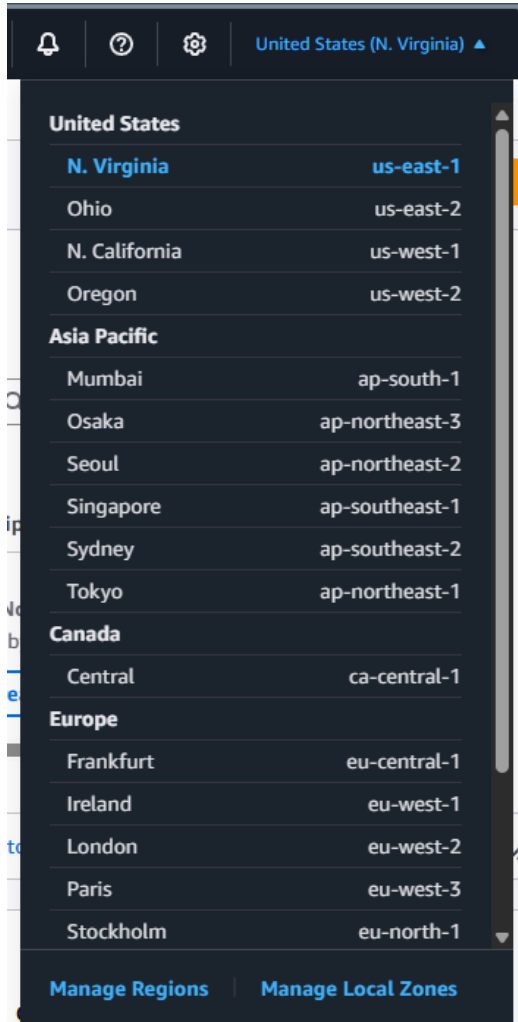


To test stop protection go to EC2, choose “Instances”, go to “Instance state” and then select “Stop instance”.

Problems:

For the First lab in the section where the user is meant to stop the instance, the instance might not be there when the user logs into the IAM user account.

To fix this issue you need to switch to the region you used at the start of the lab, as shown below:



Conclusion:

Throughout these 3 labs the user learns how to manage users, security credentials, and permissions on AWS resources all through AWS

Identity and Access Management. The labs also demonstrate how Amazon Virtual Private Clouds function and enable the user to launch their own VPC and observe it. Finally, Amazon Elastic Compute Cloud is also showcased in the final lab where the user sets up failure resilience and can test it themselves.